

Data Science Project Title

Domain-Adaptive Sentiment & Public Opinion Analysis using Transformers and Large Language Models

Project Overview

In real-world natural language processing (NLP) applications, labeled data is often scarce in specialized domains such as financial commentary, medical text, or policy analysis. Models trained on general-purpose datasets (e.g., product reviews or social media posts) typically perform poorly when applied to domain-specific text because of differences in vocabulary, tone, semantics, and contextual usage.

This project requires students to design and implement a domain-adaptive sentiment analysis system that transfers knowledge from a large general-domain dataset to a smaller specialized dataset (e.g., financial commentary). Students will explore modern NLP approaches, including Transformer architectures, transfer learning strategies, and Large Language Models (LLMs), and experimentally analyze how different training strategies affect performance.

The primary goal is not only to build a model, but to demonstrate experimentally how domain adaptation improves performance.

Learning Objectives

By completing this project, students will:

- Understand how Transformer-based models represent contextual meaning.
 - Apply transfer learning to adapt models to new domains.
 - Compare traditional ML, deep learning, and LLM-based approaches.
 - Analyze the effect of domain shift on prediction performance.
 - Evaluate models using quantitative metrics and qualitative error analysis.
 - Implement a complete NLP pipeline from preprocessing to evaluation.
 - Develop experimental reasoning skills through controlled comparisons.
-

Datasets and Their Roles

Students must use two datasets, each serving a distinct purpose in training.

Source Dataset (General Domain)

Examples: product reviews, movie reviews, tweets

Used for:

- learning general sentiment patterns
- learning language structure
- stabilizing model training
- providing large-scale training signals

This dataset is not domain-specific.

Target Dataset (Domain-Specific)

Examples: financial commentary, policy statements, industry reports

Used for:

- adapting the model to specialized vocabulary
- learning domain-specific sentiment meaning
- evaluating real-world performance

This dataset is typically smaller but more realistic.

Core Concept Students Must Demonstrate

General dataset teaches language understanding.

Domain dataset teaches domain meaning.

Controlled Experimental Models (A / B / C Framework)

To isolate the effect of training data, students must train the same model architecture under three different training conditions.

The architecture must remain identical.

Only the training data may change.

Model A — General Only (No Domain Adaptation)

Training data:

General dataset only

Testing data:

Domain dataset

Purpose:

- Measure how well general knowledge transfers to a new domain.

Expected behavior:

- understands sentiment words
 - struggles with domain terminology
-

Model B — Domain Only (No Transfer Learning)

Training data:

Domain dataset only

Testing data:

Domain dataset

Purpose:

- Measure performance when trained only on limited domain data.

Expected behavior:

- understands domain vocabulary
 - may overfit
 - unstable performance
-

Model C — Mixed Training

Training data:

General dataset + Domain dataset combined

Testing data:

Domain dataset

Purpose:

- Evaluate whether combining datasets improves performance stability.

Expected behavior:

- balanced performance
 - better generalization
-

Required Comparison Insight

Students must experimentally show:

Model Typical Behavior

- | | |
|---|---------------------------|
| A | general but domain-weak |
| B | domain-aware but unstable |
| C | most balanced |

Project Tasks

1. Data Preparation

Students must:

- clean and preprocess both datasets
- normalize text
- tokenize input
- remove noise
- perform exploratory data analysis (EDA)

Deliverable: data analysis report with statistics and visualizations.

2. Baseline Modeling (Traditional Machine Learning)

Students must train classical models such as:

- Logistic Regression
- SVM
- Naïve Bayes

using TF-IDF features.

They must train Model A, B, and C using these methods.

This establishes performance baselines.

3. Transformer-Based Modeling

Students must use a pretrained Transformer model (e.g., BERT variant).

They must implement and compare three training strategies:

Strategy 1 — Domain Only Fine-Tuning

Train only on domain dataset.

Strategy 2 — Sequential Transfer Learning

Train on general dataset → then fine-tune on domain dataset.

(This is the primary transfer learning experiment.)

Strategy 3 — Mixed Training

Train once on combined dataset.

Students must compare all strategies quantitatively.

4. Transfer Learning Techniques (Required Section)

Students must implement at least two of the following:

- Sequential fine-tuning (general → domain)
- Domain-Adaptive Pretraining (DAPT)
- Dataset weighting strategies
- Few-shot fine-tuning
- Domain tags in training text

This section explicitly evaluates students' understanding of transfer learning.

5. LLM-Based Approach

Students must evaluate sentiment classification using a Large Language Model without full training.

Required experiments:

- zero-shot prompting
- few-shot prompting
- structured prompts

Students must compare LLM performance with trained models.

6. Evaluation and Analysis

Students must report:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

They must also analyze:

- domain vocabulary errors
 - prediction failures
 - bias patterns
 - cross-domain performance gaps
-

Required Analytical Question

Students must explicitly answer:

How does training data choice influence model performance?

This must be supported by experimental results.

Final Deliverables

Each group must submit:

1. Technical report (methodology, experiments, results)
 2. Source code repository with documentation
 3. Presentation/demo
 4. Model comparison analysis
-

Expected Outcomes

Students completing this project will gain hands-on experience with:

- Transformer architectures
- Transfer learning workflows
- Prompt engineering
- Experimental evaluation design
- Domain adaptation strategies

They will produce a production-style NLP pipeline suitable for portfolio use.

Industry Relevance

This project reflects real-world machine learning challenges where labeled data is limited and domain adaptation is necessary, including:

- financial sentiment monitoring
- public opinion analysis
- brand perception tracking
- medical document classification

Students completing this project will develop skills aligned with roles in:

- Data Science
 - NLP Engineering
 - Applied AI
-

Key Concept Students Must Learn

Model performance depends not only on architecture, but on how well training data matches the target

domain.

Weekly Schedule

Tutorial 1: Introduction

Tutorial 2: AI, ML, Deep learning, Generative AI

Tutorial 3: Transformer Model

Tutorial 4: LLM, LLM reasoning

Tutorial 5: Building a Production-Ready Machine Learning Pipeline

Tutorial 6: Semi-supervised learning & Handling Data

Tutorial 7: Transfer learning

Tutorial 8: Social and cultural technology design with AI perspectives

Tutorial 9: Human in the loop with LLM

Tutorial 10: Chain of thoughts

Tutorial 11: Technical Report Writing

Tutorial 12: Wrap Up